SYSTEM DESIGN DOCUMENT

AI Weather Intelligence & Autonomous Decision Support System (WeatherPulse)

Contributors: Dushyant (@dushyant1704) | AVI192 (@AVI192) | **Repository:** github.com/dushyant1704/ai-weather-intelligence

1. System Architecture

WeatherPulse is architected as a decoupled, multi-tiered AI system comprising a responsive client dashboard (React 19), a resilient REST backend microservice (Node.js/Express 5), an autonomous Multi-Agent Reasoning Engine, and Groq LLaMA 3.3 70B Versatile inference.

2. Autonomous Agent Roles & Responsibilities

Agent Name	Role & Core Responsibilities	Key Algorithmic Output
AI Orchestrator (orchestratorAgent.js)	Natural language intent classification (ACTIVITY, RISK, COMPARISON, TREND), entity/city resolution, dynamic workflow planner, pipeline coordinator.	Execution plan, routing confidence, resolved coordinates, synthesized state envelope.
Data Intelligence Quality Gate (dataIntelligenceAgent.js)	Validates physical bounds, telemetry completeness, sensor integrity, and detects missing forecast steps.	Data Quality Score (0–100%), readiness boolean, missing field flags, normalized payload.
Risk Intelligence Agent (riskIntelligenceAgent.js)	Calculates compound indices across 5 dimensions: Heat Stress, UV Exposure, Precipitation, Wind, Visibility; identifies non-linear cross-effects.	Overall & Activity Risk scores (0–100), hazard classifications, active cross-effects list.
Forecast Trend Agent (forecastTrendAgent.js)	Evaluates 40 forecast horizon slices (5 days × 3 hrs) using moving window slopes and delta shifts.	Trajectory classification (IMPROVING, STABLE, WORSENING), peak shift times, delta metrics.
Weather Reasoning Agent (weatherReasoningAgent.js)	Bio-meteorological modeling via Humidex, Apparent Temperature, and Wind Chill equations.	Sensory comfort label (Comfortable to Very Uncomfortable), biothermal reasoning narrative.

Activity Decision Agent (activityDecisionAgent.js)	Synthesizes risk scores and trend offsets into activity-specific thresholds across 12 sports/events.	Suitability Score (0–100), Rating (EXCELLENT to UNSUITABLE), Favorable & Caution points.
Conflict Arbitration Agent (conflictArbitrationAgent.js)	Detects paradoxes (e.g., current sunny skies vs. imminent storm trend). Overrules flawed outputs and discovers alternative safe time windows.	Arbitrated score, override rationale, negotiated replan window (e.g., '14:00 - 17:00').
City Intelligence Agent (cityIntelligenceAgent.js)	Comparative benchmarking across multiple destination analyses for travel decision support.	Destination ranking, winner margins, comparative differential matrix.
Live Groq LLM Agent (groqService.js)	Infers executive natural-language summaries, clothing & gear recommendations, and powers the interactive Copilot chat.	Natural-language Executive Verdict, gear suggestions, conversational chat responses.

3. Communication Pattern & Orchestration Flow

• **Orchestrator-Mediated Blackboard Pattern:** Agents communicate via a shared memory context object managed strictly by the AI Orchestrator. Agents never write to uncontrolled global variables.

• **Serial & Parallel Pipeline Execution:** Data Quality is executed serially as a mandatory gate. Upon passing (>80%), independent analytical agents (Risk, Trend, Reasoning) run concurrently, followed by the Activity Decision and Conflict Arbitration agents.

• **Arbitration Interception:** If the Conflict Arbitration Agent detects diverging signals (e.g., current risk is LOW but 3-hour storm probability surges above 75%), it intercepts and mutates the decision state before passing to Groq LLM.

4. Technology Stack & Architectural Rationale

• **Frontend (React 19 + Vite 7):** Ultra-fast HMR, concurrent React rendering, responsive component architecture, Lucide iconography, and Recharts interactive trend visualization.

• **Backend (Node.js + Express 5 ES Modules):** Lightweight, high-throughput asynchronous I/O ideal for orchestrating simultaneous API calls and running analytical pipelines.

• **LLM Inference (Groq Cloud SDK - LLaMA 3.3 70B Versatile):** Ultra-low latency token generation (sub-second TTFT) enabling real-time conversational co-piloting and verdict synthesis without blocking the UI.

• **Telemetry Provider (OpenWeatherMap API 2.5/3.0):** Comprehensive global meteorological data covering current weather, 5-day / 3-hour forecasts, UV index, and global geocoding.